
Affinoid algebras



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1 Normed rings and modules

1.1 Semi-normed algebraic structures

Definition 1.1. Let G be an abelian group. A *semi-norm* on G is a function $\|\cdot\| : G \rightarrow \mathbb{R}_{\geq 0}$ such that

- $\|0\| = 0$;
- $\forall x, y \in G, \|x + y\| \leq \|x\| + \|y\|$.

Suppose that R is a ring (commutative with unity) and $\|\cdot\|$ is a semi-norm on the underlying abelian group of R . We further require that

- $\|1\| = 1$;
- $\forall x, y \in R, \|xy\| \leq \|x\|\|y\|$.

Suppose that $(M, \|\cdot\|_M)$ is an R -module and $\|\cdot\|_M$ is a semi-norm on the underlying abelian group of M . We further require that

- $\forall a \in R, x \in M, \|ax\|_M \leq \|a\|\|x\|_M$.

Suppose that $(A, \|\cdot\|_A)$ is an R -algebra and $\|\cdot\|_A$ is a semi-norm on the underlying R -module of A . We further require that this semi-norm is a semi-norm on the underlying ring of A .

Definition 1.2. Let A be an abelian group (or ring, R -module, R -algebra) equipped with a semi-norm $\|\cdot\|$. We say that $\|\cdot\|$ is a *norm* if $\forall x \in A, \|x\| = 0 \iff x = 0$.

Definition 1.3. Let A be an abelian group (or ring, R -module, R -algebra) equipped with a semi-norm $\|\cdot\|$. We say that $\|\cdot\|$ is *non-archimedean* if we have the strong triangle inequality $\forall x, y \in A, \|x + y\| \leq \max(\|x\|, \|y\|)$.

Definition 1.4. Let $\|\cdot\|_1$ and $\|\cdot\|_2$ be two semi-norms on an abelian group (or ring, R -module, R -algebra) A . We say $\|\cdot\|_1$ is *bounded* by $\|\cdot\|_2$ if there exists a constant $C > 0$ such that $\forall x \in A, \|x\|_1 \leq C\|x\|_2$. If $\|\cdot\|_1$ and $\|\cdot\|_2$ are bounded by each other, we say they are *equivalent*.

Remark 1.5. Equivalent semi-norms induce the same topology on A . However, the converse is not true in general. Compare with ??.

Yang: what about on a module?

Definition 1.6. Let M be a semi-normed abelian group (or R -module) and $N \subseteq M$ be a subgroup (or R -submodule). The *residue semi-norm* on the quotient group M/N is defined as

$$\|x + N\|_{M/N} = \inf_{y \in N} \|x + y\|_M.$$

Unless otherwise specified, we always equip the quotient M/N with the residue semi-norm.

Remark 1.7. The residue semi-norm is a norm if and only if N is closed in M .

Definition 1.8. Let M and N be two semi-normed abelian groups (or rings, R -modules, R -algebras). A homomorphism $f : M \rightarrow N$ is called *bounded* if there exists a constant $C > 0$ such that $\forall x \in M, \|f(x)\|_N \leq C\|x\|_M$.

A bounded homomorphism $f : M \rightarrow N$ is called *admissible* if the induced isomorphism $M/\ker f \rightarrow \operatorname{Im} f$ is an isometry, i.e., $\forall x \in M, \|f(x)\|_N = \|x\|_{M/\ker f}$.

Definition 1.9. A semi-norm $\|\cdot\|$ on a ring R is called *multiplicative* if $\forall x, y \in R, \|xy\| = \|x\|\|y\|$. It is called *power-multiplicative* if $\forall x \in R, \|x^n\| = \|x\|^n$ for all integers $n \geq 1$. A multiplicative norm sometimes is called a *(multiplicative) valuation* or an *absolute value*.

Example 1.10. Let R be arbitrary ring. The *trivial norm* on R is defined as $|x| = 0$ if $x = 0$ and $|x| = 1$ if $x \neq 0$. The ring R equipped with the trivial norm is a valuation ring. This norm is non-archimedean and multiplicative.

Example 1.11. A valuation field $(\mathbf{k}, |\cdot|)$ can be viewed as a valuation ring.

Example 1.12. Let $|\cdot| = |\cdot|_\infty$ be the usual absolute value on $\mathbb{Z}$. Then $(\mathbb{Z}, |\cdot|)$ is a valuation ring.

Example 1.13. Let X be a compact Hausdorff topological space. The ring $\mathcal{C}(X, \mathbb{R})$ of continuous real-valued functions on X equipped with the norm $\|f\| = \sup_{x \in X} |f(x)|$ is a normed $\mathbb{R}$ -algebra. Its norm is power-multiplicative but not multiplicative in general. It is worth mentioning that the Gelfand-Kolmogorov Theorem saying that we can recover X from the normed ring $\mathcal{C}(X, \mathbb{R})$.

Example 1.14. Let K be a number field and $\mathcal{O}_K$ be its ring of integers. The action of K on itself by multiplication induces an embedding $K \hookrightarrow \text{End}_{\mathbb{Q}}(K) \cong M_n(\mathbb{Q})$. Consider a norm $\|\cdot\|$ on the matrix ring $M_n(\mathbb{Q})$ and restrict it to K . Then $(K, \|\cdot\|)$ is a normed ring. This norm is even not power-multiplicative in general.

Definition 1.15. A (semi-)norm on an abelian group M induces a (pseudo-)metric $d(x, y) = \|x - y\|$ on M . A (semi-)normed abelian group M is called *complete* if it is complete as a (pseudo-)metric space.

Definition 1.16. A *banach ring* is a complete normed ring.

Proposition 1.17. Let R be a banach ring and $I \subseteq R$ be a closed ideal. Then the residue norm on the quotient ring R/I is a norm for rings.

Proof. We only need to show that $\|1\|_{R/I} = 1$. Yang: To be added. □

Proposition 1.18. Let R be a banach ring. Then the group of invertible elements $R^\times$ is an open subset of R .

Proof. Yang: To be added. □

Corollary 1.19. Let R be a banach ring. Then every maximal ideal of R is closed.

Proof. Yang: To be added. □

Definition 1.20. Let $(A, \|\cdot\|_A)$ be a normed algebraic structure, e.g., a normed abelian group, a normed ring, or a normed module. The *completion* of A , denoted by $\hat{A}$, is the completion of A as a metric space. Since A is dense in its completion and the algebraic operations are uniformly continuous, the algebraic operations on A can be uniquely extended to the completion.

Let R be a normed ring and M, N be semi-normed R -modules. There is a natural semi-norm on the tensor product $M \otimes_R N$ defined as

$$\|z\|_{M \otimes_R N} = \inf \left\{ \sum_i \|x_i\|_M \|y_i\|_N : z = \sum_i x_i \otimes y_i, x_i \in M, y_i \in N \right\}.$$

Definition 1.21. Let R be a banach ring and M, N complete semi-normed R -modules. The *complete tensor product* $M \hat{\otimes}_R N$ is defined as the completion of the semi-normed R -module $M \otimes_R N$.

Example 1.22. Yang: Example of complete tensor product.

1.2 Spectral radius

Definition 1.23. Let R be a banach ring. For each $f \in R$, the *spectral radius* of f is defined as

$$\rho(f) = \lim_{n \rightarrow \infty} \|f^n\|^{1/n}.$$

Since $\|\cdot\|$ is submultiplicative, the limit defining $\rho(f)$ exists and equals to $\inf_{n \geq 1} \|f^n\|^{1/n}$ by Fekete's Subadditive Lemma.

Example 1.24. Consider the normed ring $(K, \|\cdot\|)$ in [Example 1.14](#). Suppose that $\|\cdot\|$ on $M_n(\mathbb{Q})$ is given by the 2-norm induced by the euclidean norm on $\mathbb{Q}^n$. Then for a matrix $A \in M_n(\mathbb{Q})$, its spectral radius $\rho(A)$ is given by the largest absolute value of eigenvalues of A . Hence for each $a \in K$, its spectral radius $\rho(a)$ is given by the largest absolute value of embeddings of a into $\mathbb{C}$, i.e.,

$$\rho(a) = \max_{\sigma: K \hookrightarrow \mathbb{C}} |\sigma(a)|.$$

Proposition 1.25. Let $(R, \|\cdot\|)$ be a banach ring. The spectral radius $\rho(\cdot)$ defines a power-multiplicative semi-norm on R that is bounded by $\|\cdot\|$.

Proof. Yang: To be continued. □

Definition 1.26. A banach ring R is called *uniform* if its norm is power-multiplicative.

Definition 1.27. Let R be a banach ring. The *uniformization* of R , denoted by $R \rightarrow R^u$, is the banach ring with the universal property among all bounded homomorphisms from R to uniform banach rings. Yang: To be continued.

Definition 1.28. Let R be a banach ring. An element $f \in R$ is called *quasi-nilpotent* if $\rho(f) = 0$. All quasi-nilpotent elements of R form an ideal, denoted by $\text{Qnil}(R)$.

Proposition 1.29. Let R be a banach ring. The completion of $R/\text{Qnil}(R)$ with respect to the spectral radius $\rho(\cdot)$ is the uniformization of R .

Proof. Yang: To be continued. □

1.3 Reduction rings

Definition 1.30. Let R be a non-archimedean banach ring. We define

$$R^\circ = \{f \in R : \rho(f) \leq 1\}, \quad R^{\circ\circ} = \{f \in R : \rho(f) < 1\}.$$

The *reduction ring* of R is defined as the quotient ring

$$\tilde{R} = R^\circ / R^{\circ\circ}.$$

For a non-archimedean field $\mathbf{k}$, its reduction ring $\tilde{\mathbf{k}} = \mathcal{K}_{\mathbf{k}}$ is just the residue field of its valuation ring.

Example 1.31. Let R be a ring equipped with the trivial norm. Then we have $R^\circ = R$ and $R^{\circ\circ} = \text{nil}(R)$. Hence the reduction ring $\tilde{R}$ is isomorphic to the reduced ring $R_{\text{red}} = R/\text{nil}(R)$.

Yang: To be continued...

2 Convergent and restricted power series

Notation 2.1. Let $T = (T_1, \dots, T_n)$ be a tuple of n indeterminates, $r = (r_1, \dots, r_n)$ be a tuple of n positive real numbers, and $\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{N}^n$ be a multi-index. We use the following notations:

- $T^\alpha := T_1^{\alpha_1} T_2^{\alpha_2} \dots T_n^{\alpha_n}$ and $r^\alpha := r_1^{\alpha_1} r_2^{\alpha_2} \dots r_n^{\alpha_n}$;
- $\underline{T/r} := (T_1/r_1, T_2/r_2, \dots, T_n/r_n)$;
- $|\alpha| := \alpha_1 + \alpha_2 + \dots + \alpha_n$;

2.1 Absolutely convergent power series

Definition 2.2. Let R be a banach ring and $r > 0$ be a real number. We define the *ring of absolutely convergent power series* over R with radius r as

$$R\langle T/r \rangle := \left\{ \sum_{n=0}^{\infty} a_n T^n \in R[[T]] : \sum_{n=0}^{\infty} |a_n| r^n < \infty \right\}.$$

Equipped with the norm $\| \sum_{n=0}^{\infty} a_n T^n \| := \sum_{n=0}^{\infty} |a_n| r^n$, the ring $R\langle T/r \rangle$ is a banach ring.

For a tuple of n indeterminates $T = (T_1, \dots, T_n)$ and a tuple of n positive real numbers $r = (r_1, \dots, r_n)$, we define

$$R\langle \underline{T/r} \rangle := R\langle T_1/r_1, \dots, T_n/r_n \rangle := R\langle T_1/r, \dots, T_{n-1}/r_{n-1} \rangle \langle T_n/r_n \rangle.$$

Note that if R has trivial norm, then

$$R\langle T/r \rangle = \begin{cases} R[[T]], & \text{if } r < 1; \\ R[T], & \text{if } r \geq 1. \end{cases}$$

Proposition 2.3. Let R be a banach ring and $r > 0$ be a real number. Then the ring of absolutely convergent power series $R\langle T/r \rangle$ is a banach ring. **Yang: To be checked.**

In general, the ring of absolutely convergent power series is not even power-multiplicative. The following examples describe the uniformization of the ring of absolutely convergent power series over fields.

Example 2.4. Let $A = \mathbb{C}\langle T/r \rangle$. For $f \in A$, set

$$\|f\|_0 := \max_{x \in E(0,r)} |f(x)|.$$

I claim that $\|\cdot\|_0 = \rho_A(\cdot)$. **Yang: To be continued.**

Example 2.5. Let $\mathbb{k}$ be an algebraically closed non-archimedean complete field and $A = \mathbb{k}\langle T/r \rangle$.

Yang: To be continued.

2.2 Tate algebras

Definition 2.6. Let R be a non-archimedean banach ring. The *Tate algebra* over R with radius $r > 0$ in one variable is defined as

$$R\{T/r\} := \left\{ \sum_{n=0}^{\infty} a_n T^n \in R[[T]] : |a_n| r^n \rightarrow 0 \text{ as } n \rightarrow +\infty \right\}.$$

For a tuple of n indeterminates $T = (T_1, \dots, T_n)$ and a tuple of n positive real numbers $r = (r_1, \dots, r_n)$, we define

$$R\{\underline{T}/r\} := R\{T_1/r_1, \dots, T_n/r_n\} := R\{T_1/r_1, \dots, T_{n-1}/r_{n-1}\}\{T_n/r_n\}.$$

Elements in $R\{\underline{T}/r\}$ are also called *restricted power series*.

Proposition 2.7. Let R be a non-archimedean banach ring. Then the Tate algebra $R\{T/r\}$ is a non-archimedean multiplicative banach R -algebra with respect to the *gauss norm*

$$\left\| \sum_{n \geq 0} a_n T^n \right\| := \sup_{n \geq 0} |a_n| r^n = \max_{n \geq 0} |a_n| r^n.$$

Proof. Yang: To be added. □

The following examples describe the reduction rings of Tate algebras.

Example 2.8. Let R be a non-archimedean banach ring and $A = R\{T\}$ be the Tate algebra in one variable over R . Then we have

$$A^\circ = \left\{ \sum_{n \geq 0} a_n T^n : |a_n| \leq 1 \text{ for all } n \in \mathbb{N} \right\} = R^\circ\{T\},$$

and

$$A^{\circ\circ} = \left\{ \sum_{n \geq 0} a_n T^n : |a_n| < 1 \text{ for all } n \in \mathbb{N} \right\} = R^{\circ\circ}\{T\}.$$

Since the norm of items in a restricted power series will tend to 0, we have

$$\tilde{A} = \tilde{R}[T].$$

Example 2.9. Let R is a multiplicative non-archimedean banach ring. Set

$$\sqrt{|R|^{-1}} = \{r \in \mathbb{R}_{>0} : r^{-n} \in |R| \text{ for some } n \in \mathbb{N}_{>0}\}.$$

Fix $r \in \mathbb{R}_{>0}^n$, consider the Tate algebra $A = R\{T/r\}$.

Suppose that $r \in \sqrt{|R|^{-1}}$. Let n be the minimal positive integer such that $r^n \in |R|^{-1}$ and

$$\tilde{M}_k := \{a \in R : |a| = r^{-nk}\} / \{a \in R : |a| < r^{-nk}\}.$$

For $a_m T^m$ with $n \nmid m$, we have $\|a_m T^m\| = |a_m| r^m \leq 1 \implies |a_m| r^m < 1$. Hence

$$\widetilde{R\{T/r\}} = \tilde{R} \oplus \tilde{M}_1 T^n \oplus \tilde{M}_2 T^{2n} \oplus \tilde{M}_3 T^{3n} \oplus \dots$$

In case $R = \mathbf{k}$ is a non-archimedean field, we have $\tilde{M}_k \cong \tilde{\mathbf{k}}$ by choosing an element $c \in \mathbf{k}$ with

$|c| = r^{-n}$. Hence

$$\widehat{\mathbf{k}\{T/r\}} \cong \widehat{\mathbf{k}[T^n]}.$$

Suppose that $r \notin \sqrt{|R|^{-1}}$. Then for every $\|a_n T^n\| = a_n r^n \leq 1$, we have $|a_n| < 1$. It follows that

$$\widehat{R\{T/r\}} = \widetilde{R}.$$

Definition 2.10. Let R be a non-archimedean banach ring and $A = R\{T/r\}$. For $f = \sum_{a_n \in \mathbb{N}} a_n T^n \in A$, we define the *degree* of f as

$$\deg f := \max\{n \in \mathbb{N} : \|a_n\| r^n = \|f\|\}.$$

Easily to see that

$$\deg(f \cdot g) = \deg f + \deg g$$

and

$$\deg(f + g) \leq \max\{\deg f, \deg g\}.$$

It is interesting to note that if R has trivial norm, then $\deg f$ coincides with the usual degree of polynomials when $r \geq 1$ and the order of formal power series when $r < 1$.

Definition 2.11. Let R be a non-archimedean banach ring and $A = R\{T/r\}$. A restricted power series $f = \sum_{n \in \mathbb{N}} a_n T^n \in A$ of degree d is said to be *distinguished* if a_d is invertible in R .

Proposition 2.12. Let R be a non-archimedean banach ring. An element $f \in R\{T/r\}$ is invertible if and only if it is distinguished of degree 0.

Proof. Denote $f = \sum_{n=0}^{\infty} a_n T^n$. Multiplying by a_0^{-1} , we can reduce to the case $a_0 = 1$. Let $g = \sum_{n=0}^{\infty} b_n T^n$ be the inverse of f in $R[[T]]$. Then we have

$$f \cdot g = \sum_{k=0}^{\infty} \left(\sum_{n+m=k} a_n b_m \right) T^k = 1 \implies \forall k \geq 1, b_k = - \sum_{\substack{n+m=k \\ n \neq 0}} a_n b_m.$$

Set $A = \|f - 1\| < 1$. We show that for every $m \in \mathbb{N}$, there exists $B_m > 0$ such that for all $k \geq B_m$, we have $\|b_k\| r^k \leq A^m$. This implies that $\|b_k\| r^k \rightarrow 0$ as $k \rightarrow +\infty$ and hence $g \in R\{T/r\}$.

Note that $b_0 = 1$. By induction on k , we have

$$\|b_k\| r^k \leq \max_{\substack{n+m=k \\ n \neq 0}} \|a_n\| r^n \cdot \|b_m\| r^m \leq A \max_{m < k} \|b_m\| r^m \leq A.$$

Suppose that the claim holds for m . There exists $C_{m+1} \in \mathbb{N}$ such that for all $n \geq C_{m+1}$, we have $\|a_n\| r^n \leq A^{m+1}$. Set $B_{m+1} = B_m + C_{m+1} + 1$. For any $k \geq B_{m+1}$, we have

$$\|b_k\| r^k \leq \max_{\substack{n+m=k \\ n \neq 0}} \|a_n\| r^n \cdot \|b_m\| r^m \leq \max\{A^{m+1}, A \cdot A^m\} = A^{m+1}$$

since either $n \geq C_{m+1}$ or $m \geq B_m$. Hence the claim holds for $m + 1$. $\square$

2.3 Weierstrass preparation

Definition 2.13. Let R be a non-archimedean banach ring and $A = R\{T/r\}$. A Weierstrass polynomial is a monic polynomial $P \in A[T] \subset R\{T/r\}$ whose two degrees as a polynomial and as a restricted power series coincide.

Theorem 2.14 (Weierstrass preparation theorem). Let R be a non-archimedean banach ring. Let $f \in R\{T/r\}$ be a distinguished restricted power series of degree d . Then there exists a unique Weierstrass polynomial $p \in R[T]$ of degree d and a unique unit $u \in R\{T/r\}$ such that

$$f = p \cdot u.$$

Yang: To be checked.

Proof. Yang: To be added. □

Remark 2.15. In my knowledge, there are at least three different versions of Weierstrass preparation theorem under different settings:

- The classical Weierstrass preparation in complex analysis;
- The Weierstrass preparation for formal power series over complete noetherian local rings;
- The Weierstrass preparation for Tate algebras over non-archimedean banach rings.

Let $(R, \mathfrak{m})$ be a complete noetherian local ring. Note that there is also a Weierstrass preparation theorem for formal power series over R stating that for every formal power series $f \in R[[T]]$ whose reduction $\bar{f} \in (R/\mathfrak{m})[[T]]$ is of order d , there exists a unique monic polynomial $p \in R[T]$ of degree d and a unique unit $u \in R[[T]]$ such that

$$p \equiv T^d \pmod{\mathfrak{m}}, \quad f = p \cdot u.$$

Yang: To be continued.

Corollary 2.16. Let $\mathbf{k}$ be a complete non-archimedean field and $r > 0$ be a positive real number. Then the Tate algebra $\mathbf{k}\{T/r\}$ is an euclidean domain with respect to the degree defined in Definition 2.10. Yang: To be added.

Proof. Let $f, g \in \mathbf{k}\{T/r\}$ be two elements with $g \neq 0$. We need to find $q, s \in \mathbf{k}\{T/r\}$ such that

$$f = q \cdot g + s, \quad \deg s < \deg g.$$

Yang: To be added. □

3 Affinoid algebras

Definition 3.1. Let R be a non-archimedean banach ring. A banach R -algebra A is called a R -*affinoid algebra* if there exists an admissible surjective homomorphism

$$\varphi : R\{\underline{T}/\underline{r}\} \twoheadrightarrow A$$

for some $\underline{r} = (r_1, \dots, r_n) \in R_{>0}^n$.

If one can choose $r_1 = \dots = r_n = 1$, then we say that A is a *strictly R -affinoid algebra*.

Example 3.2. Suppose that the norm on R is trivial. Then a strictly R -affinoid algebra is just an R -algebra of finite type equipped with the trivial norm.

Example 3.3. Let A be an R -affinoid algebra, $f_i, g_j \in A$ and $r_i, s_j \in R_{>0}$. Define

$$\begin{aligned} A\{\underline{f}/\underline{r}\} &:= A\{\underline{T}/\underline{r}\}/(T_i - f_i), \\ A\{\underline{f}/\underline{r}, \underline{s}/\underline{g}\} &:= A\{\underline{T}/\underline{r}, \underline{S}/\underline{s}\}/(T_i - f_i, g_j S_j - 1). \end{aligned}$$

Suppose that $\underline{g} \in R$ with $(f_i, \underline{g}) = A$. Then we can define

$$A\{(\underline{f}/\underline{g})/\underline{r}\} := A\{\underline{T}/\underline{r}\}/(gT_i - f_i).$$

All of the above three algebras are again R -affinoid algebras. They can be viewed as normed analogues of localizations. **Yang: To be replaced.**

In the rest of this section, we fix a complete non-archimedean field $\mathbf{k}$ and consider $\mathbf{k}$ -affinoid algebras.

3.1 Strict case

First we consider strictly $\mathbf{k}$ -affinoid algebras, which are well-studied in the era of rigid analytic geometry.

Proposition 3.4. Let $\mathbf{k}$ be a complete non-archimedean field and $f \in \mathbf{k}\{\underline{T}\}$. Then there exists an automorphism φ of $\mathbf{k}\{\underline{T}\}$ over $\mathbf{k}$ such that $\varphi(f)$ is T_n -distinguished, i.e., $\varphi(f) \in A\{T_n\}$ is distinguished in the variable T_n with $A = \mathbf{k}\{T_1, \dots, T_{n-1}\}$.

Proof. **Yang: To be added.** □

Proposition 3.5. Let $\mathbf{k}$ be a complete non-archimedean field. Then the Tate algebra $\mathbf{k}\{\underline{T}\}$ is noetherian, factorial, and Jacobson.

Proof. **Yang: To be completed.** □

Corollary 3.6. Strictly $\mathbf{k}$ -affinoid algebras are noetherian.

Proof. **Yang: To be completed.** □

Theorem 3.7. Let A be a strictly $\mathbf{k}$ -affinoid algebra. Then there exists a finite injective admissible

homomorphism

$$\varphi : \mathbf{k}\{T_1, \dots, T_d\} \hookrightarrow A.$$

Proof. Yang: To be completed. □

Proposition 3.8. Let A be an $\mathbf{k}$ -affinoid algebra. Then there exists a constant $C > 0$ and $N > 0$ such that for all $f \in A$ and $n \geq N$, we have

$$\|f^n\| \leq C\rho(f)^n.$$

In particular, $\text{Qnil}(A) = \text{nil}(A)$.

Furthermore, if A is reduced, we have

$$\|f\| \leq C\rho(f)$$

for all $f \in A$.

Proof. Yang: To be completed. □

3.2 General case

Proposition 3.9. Let A be an affinoid $\mathbf{k}$ -algebra. If and only if $\rho(f) \in \sqrt{|\mathbf{k}|} \cup \{0\}$ for all $f \in A$, then A is strict. Yang: To be complete.

Proof. Yang: To be completed. □

Definition 3.10. Let $\mathbf{k}$ be a non-archimedean field. We define the *ring of restricted Laurent series* over $\mathbf{k}$ as

$$\mathbf{K}_r = \mathbf{L}_{\mathbf{k},r} := \mathbf{k}\{T/r, r/T\}.$$

Yang: Is $\mathbf{K}_r$ always a field? Yang: Do we have $\mathbf{L}_{\mathbf{k},r} = \text{Frac}(\mathbf{k}\{T/r\})$?

Proposition 3.11. Let $\mathbf{k}$ be a non-archimedean field. If $r \notin \sqrt{|\mathbf{k}^\times|}$, then $\mathbf{K}_r$ is a complete non-archimedean field with non-trivial absolute value extending that of $\mathbf{k}$.

Yang: Tensor with $\mathbf{K}_r$.

Proposition 3.12. Let A be a $\mathbf{k}$ -affinoid algebra. Then there exists $r_i \in \mathbb{R}_{>0}$ such that

$$\mathbf{K}_{r_i} \hat{\otimes}_{\mathbf{k}} A$$

is a strictly $\mathbf{K}_{r_i}$ -affinoid algebra.

There are three different categories of finite modules over an affinoid algebra A :

- The category $\mathbf{Banmod}_A$ of finite banach A -modules with A -linear maps as morphisms.
- The category $\mathbf{Banmod}_A^b$ of finite banach A -modules with bounded A -linear maps as morphisms.
- The category $\mathbf{mod}_A$ of finite A -modules with all A -linear maps as morphisms.

Theorem 3.13. Let A be an affinoid $\mathbf{k}$ -algebra. Then the category of finite banach A -modules with bounded A -linear maps as morphisms is equivalent to the category of finite A -modules with A -linear maps as morphisms. Yang: To be revised.

For simplicity, we will just write $\text{mod } A$ to denote the category of finite banach A -modules with bounded A -linear maps as morphisms.

DRAFT
